8 November 1974
SUPERSEDING
MIL-I-7566B(ASG)
2 April 1962

MILITARY SPECIFICATION

INDICATOR SYSTEM, FUEL FLOWMETER, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for aircraft-type fuel flow indicating systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-Z-325	Zinc Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Box, Fiberboard

Military

MIL-C-17	Cables, Radio Frequency, Coaxial, .086 Semirigid, 50 ohms, RG-405/U
MIL-W-76	Wire and Cable, Hook-Up, Electrical, Insulated
MIL-P-116	Preservation Methods Of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 91/96, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-P-6906	Plates, Identification, Aircraft

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MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-25142	Luminescent Material, Fluorescent

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U S Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection Of
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics Measurement Of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization Of
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing Of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimilar Metals
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form Of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Component parts. Each single indication system shall consist of one flow transmitter, one flow-rate indicator, and an intermediate device, where required.

3.2.1 Each multijet-engine system, unless otherwise specified, shall consist of a transmitter and an individual flow-rate indicator for each engine and one total-rate indicator with or without fuel remaining for the entire aircraft.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose. The use of lightweight materials and weight-saving designs shall be a major consideration, and their use shall be investigated and exploited to the greatest possible extent.

3.3.1 Metals. Metals shall be of the corrosion-resistant type or suitable treated to resist corrosion in the presence of fuels, exhaust gas, salt spray, or atmospheric conditions to which the equipment will be subjected when in storage or during normal service use.

*3.3.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined by MIL-STD-889.

3.3.3 Nonmagnetic materials. Nonmagnetic material shall be used for all parts, except where magnetic materials are essential.

3.3.4 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be protected or treated with a fungicidal agent as approved by the procuring activity, or shall be protected by hermetic sealing.

3.3.5 Corrosive fumes. Materials used in the interior construction of the system components shall not produce corrosive or deleterious fumes under the test conditions specified herein.

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3.3.6 Protective treatment. When materials are used in the construction of the system components that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.7 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Standard MIL-STD-143, except as provided in the following paragraph.

*3.3.7.1 Standard parts. Standard parts (MS and AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS and AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is not suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.4 Design and construction. The fuel flow indicating system shall be the gravimetric type in which the flow rate is indicated in pounds per unit time.

*3.4.1 Unless otherwise specified, the transmitter for multiengine aircraft shall provide the signal output for both the individual and the total rate and fuel remaining or used indication.

*3.4.2 The system shall be designed for maximum accuracy and readability under high-altitude engine starting conditions and under high-altitude cruising conditions.

*3.4.3 The system components will be so constructed that no parts will work loose in service. They shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

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3.4.4 Pivots, bearings, and gears shall neither bind nor shake and shall be as nearly frictionless as practicable.

3.4.5 The components shall be so constructed that adjustments and repairs can be easily made by the personnel of overhaul bases.

3.4.6 Electrical power and wiring.

*3.4.6.1 Input electrical power. The indicating system shall be designed to operate from a 115-volt (V), 400-cycle, single-phase alternating current (ac) power supply or a 28V direct current (dc) power supply or any approved combination thereof in accordance with the characteristics specified in Standard MIL-STD-704.

*3.4.6.2 Electrical cable. Hook-up wire and external unshielded connector cable shall be in accordance with Specification MIL-W-76. Coaxial cable, if used, shall be in accordance with Specification MIL-C-17. Internal wiring shall be neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be tied to the terminals prior to soldering, where practicable.

*3.4.6.2.1 Cable length. Accuracy and sensitivity of the indicator system shall not be adversely affected by increasing or decreasing the interconnecting cables between the transmitter unit and intermediate device to a total length of 200 feet, and between the intermediate device and the indicator to a total length of 30 feet.

3.4.6.3 Electrical connectors. The electrical connectors used shall be in accordance with Specification MIL-C-5015, except where glass seal receptacles are approved for use by the procuring activity.

3.4.7 Filling medium. For hermetically sealed units, the case shall be filled with 100 percent helium or a mixture of 90 $\pm$ 2 percent nitrogen and the balance helium. The gas shall be at least 98 percent pure, free from dust particles, and contain not more than 0.006 milligram of water vapor per liter (dew-point -65° C) at the filling pressure. The absolute pressure inside the unit shall be either one atmosphere or one-half atmosphere.

3.4.8 Transmitter unit. The transmitter shall measure the fuel flow rate on a gravimetric basis without separate and distinct density correcting means.

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3.4.8.1 Range. The transmitter shall measure flow rates in the range specified in the detail specification.

3.4.8.2 Fuel. The transmitter shall be designed to operate satisfactorily with fuels in accordance with Specification MIL-G-5572 and MIL-T-5624.

3.4.8.3 Operation pressure. The operating and test pressure of the transmitter shall be as specified in the detail specification.

3.4.8.4 Fuel line connection. Fuel line connection shall be as specified in the detail specification. The connection shall be furnished with suitable plugs or caps to prevent ingress of foreign material during storage and shipment.

3.4.8.5 Rust inhibitor. A rust inhibitor of a quality and type acceptable to the procuring activity shall be placed in the metering chamber of each transmitter, prior to sealing for shipment.

3.4.8.6 Mounting. Unless otherwise specified, the transmitter shall be designed for direct mounting on the engine.

3.4.8.6.1 Engine-mounted transmitters shall be designed to withstand an ambient operating temperature range from -55° to +121° C with fuel flowing through the transmitter and exposure to an ambient temperature of +232° C for 10-minute cycles with no fuel flow. Transmitters mounted in other positions on the aircraft shall be designed to operate over an ambient temperature range from -55° to +70° C.

3.4.8.7 Damping. The transmitter shall be provided with suitable damping means.

3.4.8.8 The transmitter body screws shall be safetied with lock wire. Not more than three screws shall be safetied with one length of wire. The lock wire shall be twisted tight between each pair of screws.

3.4.9 Indicators.

3.4.9.1 The indicator case design shall be in accordance with Standard MS33639, as specified in the detail specification.

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3.4.9.2 Hermetically sealed cases. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The indicator shall be so designed that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixture, unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected.

3.4.9.3 Cover glass. The cover glass shall be suitable for hermetic sealing and shall be clear, flat, and free from flaws which interfere with the normal reading of the instrument. The glass shall not be fluorescent when it is exposed to ultraviolet light rays in the 365-millimicron region.

3.4.9.4 Dials. The dials shall be of metal and shall be securely attached to the frame of the indicator by at least two screws in such a manner that it will not turn or loosen when the indicator is vibrated.

3.4.9.5 Dial markings. The styles and proportions of numerals and letters placed on the dial shall conform to Standard MS33558. Numerals shall distinctly indicate the graduation to which each applies. Markings and the pointer shall be finished in fluorescent-luminescent material in accordance with Specification MIL-L-25142, or lusterless white Color Number 37875 of Federal Standard No. 595, as specified in the detail specification.

3.4.9.6 Black surfaces. All surfaces and markings not otherwise specified shall be finished in lusterless black, Color No. 37038 of Federal Standard No. 595.

3.4.9.7 Pointer. The indicator pointer shall be the type specified in the detail specification. The pointer shall be light and sufficiently rigid to prevent oscillation under vibration. The pointer length and overlap shall be as specified in Standard MS33585.

3.4.10 Intermediate device. The intermediate device shall include all electrical parts of the system exclusive of those used in the transmitter and the indicator. The design of the intermediate device shall be as compact as practical.

3.4.10.1 Chassis. The chassis for electronic devices such as amplifiers shall be of metal and shall be designed to form a rigid support for the equipment mounted thereon.

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3.4.10.2 Enclosures. Amplifiers, bridge units, or other electronic assemblies shall be enclosed and sealed sufficiently to protect the internal parts from changeable climatic and atmospheric conditions and to ensure performance as specified under 3.9.

3.4.10.3 Mounting bases. Each separate component of the intermediate device shall be provided with a suitable mounting base.

3.4.10.4 Vibration and shock insulation. Delicate devices such as vacuum tubes, or other parts whose life or performance would be adversely affected by shock and vibration, shall be protected from shock and vibration occurring in any direction either by the mounting of the article of equipment itself or by local cushioning of such devices. If cushioning is employed, sufficient clearance shall be provided between parts to preclude the possibility of a cushioned part striking any other part under conditions of a violent shock.

3.4.11 Electrical parts. All standard parts, such as resistors, condensers, and transformers shall conform to Specification MIL-E-5400. Parts not conforming to Specification MIL-E-5400 may be used on specific prior approval of the procuring activity.

3.4.12 Electron tubes. Electron tubes used should be approved ruggedized construction, as listed in the preferred list of electron tubes. (see 6.5.)

*3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-1000.

3.6 Screw threads. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to Specification MIL-S-7742.

3.7 Lubrication. Lubrication shall be accomplished in accordance with Standard MIL-STD-838. Lubricants shall conform to applicable Government specifications, unless otherwise specified by the procuring activity.

3.8 Finishes and protective coatings.

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3.8.1 Finishes and protective coating which will crack, chip, or scale during normal service life or owing to extremes of atmospheric conditions, shall not be used.

3.8.2 Aluminum-alloy parts. Aluminum-alloy parts, where practicable, shall be covered with an anodic film conforming to Specification MIL-A-8625.

*3.8.3 Steel parts. Steel parts shall be cadmium or zinc plated where practicable. Cadmium plating shall be type II or III, in accordance with Specification QQ-P-416, as applicable, and of a class that is adequate to achieve the degree of protection required. Zinc plating shall be in accordance with Specification QQ-Z-325.

3.9 Performance. The indicating system or components thereof shall satisfactorily pass the tests specified in Section 4.

3.10 Identification of product.

3.10.1 Nameplate. Nameplates conforming to Specification MIL-P-6906 shall be securely attached to the transmitter unit, the intermediate device, and the indicator. The information marked in the spaces provided on the nameplate shall be in accordance with Standard MIL-STD-130 and as specified in the detail specification.

3.10.2 Use of MS, AN, or MIL designations. MS, AN, or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification.

3.11 Workmanship.

3.11.1 General. The indicator system, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.11.2 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw, bolt, or threads.

3.11.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

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3.11.4 Gears. Gear assemblies shall be properly alined and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required, gear assemblies shall be free from backlash.

3.11.5 Cleaning. All parts shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

(a) Qualification tests (4.4)
(b) Quality Conformance Inspection (4.5)

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Tapping. Except where otherwise specified, the system components shall be gently tapped or vibrated before a test reading is taken. An electrical vibrator set at 3,600 cycles per second (cps) with a maximum amplitude of 0.002 inch when attached to the test item, may be used.

4.3.3 Test position. Unless otherwise specified, the units comprising the instrument systems shall be tested in their normal operating position.

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4.3.4 Test power. Unless otherwise specified, the supply power shall be 28.0 ± 0.5V dc or 115V, 400-cycle single-phase ac. The voltage and frequency tolerance of the ac supply shall be ± 2 percent.

*4.3.5 Test fuel. Unless otherwise specified, the fuel used in tests or for calibration purposes shall conform to Specification MIL-T-5624. The specific gravity of the fuel shall be 0.785 ± 0.005 at 15.6°C.

4.3.6 Test equipment standards.

4.3.6.1 Master indicator. The master indicator shall be a standard especially designed for the purpose. The master indicator shall have a maximum error of 0.05 circular degrees or be so calibrated that the error may be subtracted from the test readings.

4.3.6.2 Master amplifier. Where an amplifier is required to operate the master indicator, the master amplifier shall be a standard especially designed for the purpose. It shall amplify the output signal from the transmitter to energize the positioning motor in the master indicator.

4.3.6.3 Master transmitter. Indicators and amplifier-indicator combinations shall be tested with a master transmitter especially designed for the purpose. The master transmitter shall provide a signal that does not have an error in excess of 0.05 circular degrees. For totalizing indication, the transmitter shall be provided with outputs equivalent to 1 to 8 transmitters. The transmitter master shall be of a design approved by the procuring activity.

4.4 Qualification tests.

4.4.1 Sampling instructions. Qualification test samples shall consist of two complete fuel flow indicator systems including all components, except where qualification is desired on components of a previously qualified system. Samples shall be identified as required and forwarded to the activity responsible for qualification, designated in the letter of authorization from that activity. (see 6.3)

4.4.1.1 Design data for qualification tests. The manufacturer shall furnish to the procuring activity, two copies of all design and calibration data required in the testing of the system and components. The data shall include, but not be limited to:

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(a) The specific gravity of the fuel used for basic calibration.
(b) Electrical values required for individually testing the transmitter unit, indicator unit, intermediate device, or specified combinations thereof.
(c) Design values used in the indicator, transmitter, intermediate device, and components thereof.
(d) Wiring diagrams as required.

4.4.2 Tests. The qualification tests of indicators shall consist of all the tests specified under quality conformance tests, and shall be conducted in the following sequence:

(a) Individual Tests
(b) Sampling Plan A Tests
(c) Sampling Plan B Tests

Qualification tests may, at the option of the activity responsible for qualification, be supplemented by actual installation and flight tests in aircraft.

*4.5 Quality conformance inspection. Quality conformance inspections shall consist of individual tests (4.5.1), sampling tests (4.5.2).

4.5.1 Individual tests. Each instrument shall be subjected to the following tests, as specified in the detail specification.

4.5.1.1 Examination of product. The instrument shall be inspected to determine compliance with this specification with respect to marking, material, workmanship, finish, proper operation, and performance.

4.5.1.2 Dielectric strength. A potential of 500V at a commercial frequency shall be applied between any pin of the electrical receptacle and the case of the indicator, before it is sealed, for a minimum period of 5 seconds. After filling and sealing, the potential shall be reduced to 200V and the same test conducted. There shall be no breakdown of the insulation.

4.5.1.3 Electrical zero (synchro-style indicators). For the purpose of this test, a single indicator with the glass removed shall be connected to the indicator being tested and used as the transmitter unit. The standard transmitter shall not be used. With the power source disconnected, the pointer of the single indicator shall be displaced 180 degrees from the pointer position of the indicator being tested. The pointer of the single indicator shall be held in this position and power shall be applied. This procedure shall be repeated a minimum of three times, after which electrical zero shall be imposed on the indicator being tested. The tip of the pointer shall be within

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0.010 inch of the electrical zero reference mark, and for dual indicators shall be within 0.010 inch of each other.

4.5.1.4 Friction and scale error.

4.5.1.4.1 Indicator. The knob of a standard transmitter shall be rotated to bring its pointer to the desired indication. Two indicator readings shall be taken; the first before, and the second after the instrument is tapped. The difference between the first and second readings shall not exceed the friction tolerance specified in the detail specification. The second reading shall not exceed the scale error specified in the detail specification.

4.5.1.4.2 Transmitter. Scale error of the transmitter shall be determined at the test points specified in the detail specification with both increasing and decreasing rates of flow. This test shall be conducted with the transmitter connected to the master indicator and the power supply. The flow with an increasing rate shall be measured by bringing the flow up to but not exceeding the specified rate. With decreasing rate, the flow shall be brought down to, but not below the specified rate. The difference between the upscale and downscale readings shall not exceed the variation tolerance specified in the detail specification. However, in no event shall either the upscale or downscale readings exceed the scale error tolerance specified in the detail specification.

4.5.1.5 Sealing (hermetically sealed instruments only). A helium leak detector or other method of test may be used upon approval by the procuring activity. When a helium leak detector is employed for conducting this test, the leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of approximately one atmosphere for the 100 percent helium filling mixture. When the helium-nitrogen mixture is used as the filling medium, the leak rate, as tested with the helium leak detector, shall not exceed 0.01 micron cubic foot per hour at a pressure differential of approximately one atmosphere.

4.5.1.6 Leak test. The transmitter shall be tested for leaks by applying a nonshock hydraulic pressure to the inlet connection with the outlet connection bled free of air and capped. The pressure shall be rapidly applied five times at room temperature. After the last application of pressure, there shall be no indication of leaks for a period of 5 minutes at the test pressure, as shown by a pressure gage readable to 10 pounds per square inch (psi), or as shown by visual leakage of a colored test fluid. The temperature of the transmitter shall then be reduced to -54° ± 2°C by submersion in a cold liquid bath

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already at this temperature for a period of not less than 15 minutes. While at this temperature, the test pressure shall be rapidly applied 5 times. After the last application of pressure, there shall be no evidence of leaks for a period of 5 minutes as detected by the same means used for room temperature portion of this test. Test pressures listed in table I shall be applied on the basis of the transmitter's design.

TABLE I

Test Pressures

Indicator type	Test pressures (psi)	
	Room temp.	-54°C
High pressure type	1,600	800
Low pressure type	200	200

4.5.1.6.1 An acceptable alternate method of cooling the transmitter instead of using a liquid bath, is to allow the transmitter to stand in an ambient air temperature of -55° ± 2°C for a period of not less than 3 hours prior to performing the leak test.

4.5.2 Sampling tests.

4.5.2.1 Sampling plan A. The number of test units shown below, depending upon the total number of items on contract, shall be selected at random and subjected to the following tests:

Total items on contract	Test units required
1 to 10	1
11 to 100	2
101 to 200	3
201 to 500	5

4.5.2.1.1 Voltage and frequency variation effect. The applied voltage for the tests shall be set at the following points:

22-1/2 volts dc	24 volts dc
105 volts ac	105 volts ac
380 cps	420 cps
24 volts dc	30 volts dc
120 volts ac	120 volts ac
380 cps	420 cps

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4.5.2.1.1.1 When the applied voltages and frequencies are changed as specified, the errors in indication shall not exceed the tolerance specified in the detail specification.

4.5.2.1.2 Power consumption. With nominal supply voltage applied to the instrument, the power consumption shall not exceed the maximum value specified in the detail specification.

4.5.2.1.3 Low temperature test. The instrument shall be placed within a test chamber and maintained at a temperature of -55° ± 2°C for a minimum period of 4 hours, and shall then be tested at this temperature for friction and scale error tests (4.5.1.4). The errors shall not exceed the tolerances specified in the detail specification.

4.5.2.1.4 High temperature test. The instrument while operating shall be subjected to a temperature of +70° ± 2°C for a period of 4 hours and shall then be tested at this temperature for friction and scale error tests (4.5.1.4). The errors shall not exceed the tolerances specified in the detail specification. The test temperature for high temperature components shall be as specified in the detail specification.

4.5.2.1.5 Vibration. The vibration test shall be conducted on the components of the instrument as specified below.

4.5.2.1.5.1 Indicator vibration resonance. The indicator amplifier combination or indicator and intermediate device shall be electrically connected to each other and to a master test instrument. The indicator shall be subjected to vibration with a circle diameter of 0.009 to 0.011 inch in a plane inclined 45 degrees to the horizontal plane at frequencies from 300 to 3,000 cycles per minute (cpm). The frequency of applied vibration shall be varied slowly throughout the range specified. The maximum double amplitude of the pointer oscillation, and the pointer variation from its original position shall not exceed the tolerances specified in the detail specification. The test shall be conducted at an indicator reading of a minimum of two points, as selected by the inspector.

4.5.2.1.5.2 Indicator vibrating failure. The indicator or indicator combination shall be electrically connected, as specified in the indicator vibration resonance test (4.5.2.1.5.1) "midscale" by means of the master test instrument. The indicator shall be subjected to vibration with a circle diameter of 0.018 to 0.020 inch in a plane inclined 45 degrees to the horizontal plane, and the frequency of vibration shall be varied uniformly from 300 to 3,000 cpm and return once each hour for a 3-hour period. Following the 3-hour period, a scale reading shall be established by means of the master test instrument and shall not exceed the tolerances specified in the detail speci-

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fication. The indicator shall be inspected thoroughly for damage or defects resulting from the vibration test.

4.5.2.1.5.3 Vibration isolation suitability-intermediate device with external shock mounts. The intermediate device, and its associated shock mounts shall be vibrated at ambient room temperature in each of 3 mutually perpendicular directions over a frequency range of 300 to 3,300 cpm at an amplitude of 0.018 inch (0.036-inch total excursion). The frequency shall be varied slowly in each direction over the entire frequency range to determine the resonant frequencies. The amplitude of the intermediate device shall be measured at the point of maximum amplitude in each direction while the frequency is varied from 1,560 to 3,300 cpm. The resonant frequencies shall be 900 cpm, or less, and efficiency of the shock mount shall be at least 65 percent at frequencies above 1,560 cpm.

4.5.2.1.5.4 Intermediate device and transmitter vibration. The intermediate device and transmitter shall be subjected to the frequency range at vibratory accelerations or double amplitudes not exceeding those shown below:

Frequency in cpm	Double amplitude or acceleration
300 to 600	0.050 inch
400 to 4,500	0.036 inch
4,500 to 30,000	±10 gravity (g)

4.5.2.1.5.4.1 Transmitter vibration resonance. The indicator, intermediate device, and transmitter, shall be electrically connected. The transmitter shall be mounted on the vibration test stand, and the indicator shall be adjusted to read approximately midscale. The resonant frequency of the transmitter shall be determined by varying the frequency of applied vibration slowly through the range of frequencies at vibratory acceleration not exceeding those specified. The test specimen shall be vibrated at the indicated resonant conditions for 1 hour (time specified refers to 1 axis of vibration) and with the applied double amplitude or vibratory accelerations specified in 4.5.2.1.5.4. These periods of vibration shall be accomplished in sequence for each of three mutually perpendicular axes of vibration applied along any 1 axis, the test period shall be conducted at the most severe resonance frequency. When resonant frequencies are not apparent within the specified frequency range, the specimen shall be vibrated twice as long as those specified for resonance at a frequency of 3,300 cpm and an applied double amplitude of 0.060 inch. While the vibration is being conducted, the maximum double amplitude of the indicator pointer oscillation and the pointer variation from its original position shall be noted to determine that they do not exceed the tolerances speci-

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fied in the detail specification. At the end of the test period, the transmitter shall be inspected thoroughly to determine that no damage or defects have resulted due to the test.

4.5.2.1.5.4.2 Transmitter vibration cycling. The transmitter shall be mounted and connected as specified in 4.5.2.1.5.4.1 and shall be vibrated with the frequency cycling between 600 and 30,000 cpm in 15-minute cycles, at an applied double amplitude of 0.036 inch, or an applied acceleration of ±10g, whichever is the limiting value. The vibration shall be conducted 1 hour in each direction along three mutually perpendicular axes. After the vibration test, a scale error test shall be conducted, and the readings established shall not exceed the room temperature tolerances specified in the detail specification. The test specimen shall be thoroughly examined to determine that no damage has resulted due to the test.

4.5.2.1.5.4.3 Intermediate device vibration resonance. The indicator, intermediate device, and transmitter shall be electrically connected. The intermediate device shall be mounted on the vibration test stand, and the indicator shall be adjusted to read approximately midscale. The resonant frequency of the intermediate device shall be determined by varying the frequency of applied vibration slowly through the range of frequencies at vibratory acceleration not exceeding those specified. The test specimen shall be vibrated at the indicated resonant conditions for 1 hour (time specified refers to 1 axis of vibration) and with the applied double amplitude or vibratory accelerations specified in 4.5.2.1.5.4. These periods of vibration shall be accomplished in sequence for each of three mutually perpendicular axes of vibration applied along any one axis. The test period shall be conducted at the most severe resonant frequency. When resonant frequencies are not apparent within the specified frequency range, the specimen shall be vibrated twice as long as specified for resonance at a frequency of 3,300 cpm and an applied double amplitude of 0.060 inch. While the vibration is being conducted, the maximum double amplitude of the indicator pointer oscillation and the pointer variation from its original position shall be noted to determine that they do not exceed the tolerances specified in the detail specification. At the end of the test period, the intermediate device shall be inspected to determine that no damage has occurred.

4.5.2.1.5.4.4 Intermediate device vibration cycling. The intermediate device shall be mounted and connected as specified in 4.5.2.1.5.4.3 and shall be vibrated with the frequency cycling between 600 and 30,000 cpm in 15-minute cycles, at an applied double amplitude of 0.036 inch, or an applied acceleration of ±10g, whichever is the limiting value. The vibration shall be conducted 1 hour in each direction along three mutually perpendicular axes. After the vibration test, an operation test shall be conducted, and the readings shall not exceed the room

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temperature tolerances specified in the detail specification. The test specimen shall be thoroughly examined for damage resulting from the test.

4.5.2.1.5.4.5 In the event that vibration testing equipment is not available to perform the cycling test as specified in 4.5.2.1.5.4.2, the cycling shall be accomplished only at frequencies ranging from 600 to 3,300 cpm and at an applied double amplitude of 0.036 inch. The frequency of vibration shall be varied uniformly from 600 to 3,300 cpm and return once each hour. The time of the vibration shall be 1 hour in each direction along three mutually perpendicular axes.

4.5.2.1.6 Position error. Recording of the 1,500 and 8,000 pph indicated values shall be made with the test instrument in the normal position. Readings obtained with the test instrument turned 90 degrees in any specified direction shall not exceed the tolerances specified in the detail specification.

4.5.2.1.7 Transmitter pressure drop. The pressure drop for the transmitter shall be determined for the rates of flow specified in the detail specification, for both normal and jammed element conditions, and shall not exceed the applicable tolerances.

4.5.2.2 Sampling plan B. Four of the inspection samples representing the manufacturer's contract or order shall be subjected to the following tests. These test samples shall be selected at random from the first 25 articles produced on the order.

4.5.2.2.1 Environmental tests. Environmental tests shall be in accordance with the specified procedures of Specification MIL-E-5272, and as specified herein.

4.5.2.2.2 Electrical characteristics. Sufficient electrical tests shall be performed by the manufacturer on the instrument and its components to insure satisfactory performance when connected in a complete flowmeter system as specified in Section 3 and in the detail specification.

4.5.2.2.3 Temperature-altitude. The temperature-altitude test for nonhermetically sealed system components only, shall be in accordance with Procedure II for a continuous period of 4 hours at -54° ± 2°C and 65,000 feet pressure altitude. While at this pressure and temperature, the instrument shall be tested for scale errors as specified in 4.5.2.2.4. The same tolerances shall apply. No damage or deterioration shall result from this test. This test may be combined with the test specified in 4.5.2.2.4.

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4.5.2.2.4 Low temperature exposure: Procedure II. Upon completion of the 24-hour period and while still at the test temperature, the instrument shall be subjected to the friction and scale error tests, and the tolerances specified in 4.5.2.1.3 shall apply. The instrument shall then be maintained at room temperature for a period of not less than 4 hours and shall then be subjected to the test specified in 4.5.1.4. The scale errors observed shall not exceed the tolerances specified for the individual tests. The instrument shall be examined for deterioration or damage caused by extreme temperature exposure.

4.5.2.2.5 High temperature exposure: Procedure II. The test temperature for high temperature components shall be as specified in the detail specification. The instrument shall then be tested at this temperature for friction and scale errors as specified in 4.5.2.1.4 and shall not exceed the amount specified therein. The instrument shall then be maintained at room temperature for a period of not less than 4 hours, and then subjected to the friction and scale error test at room temperature (4.5.1.4). The scale errors and friction observed for pointer or counter shall not exceed the tolerances specified. The instrument shall be examined for deterioration or damage caused by extreme temperature exposure.

4.5.2.2.6 Humidity: Procedure I. External connections shall be made to the instrument to simulate installed conditions. Precautions shall be taken to prevent moisture dripping on the instrument. Immediately following the completion of the fifteenth cycle, the instrument shall be subjected to the friction and scale error tests specified in 4.5.1.4. The scale errors and friction shall not exceed the tolerances specified for the individual tests. This test is not applicable for instruments contained in a hermetically sealed case.

4.5.2.2.7 Fungus: Procedure I. At the end of the test period, the equipment shall be carefully examined to ascertain that no fungus growth has occurred. This test is not applicable to completely hermetically sealed components.

4.5.2.2.8 Salt spray: Procedure I. No deterioration or damage shall result from this test. Sealed components shall be tested for leakage after this test.

* 4.5.2.2.9 Radio noise. The radio noise suppression test shall be in accordance with Standard MIL-STD-461 and MIL-STD-462, except that a frequency range of 0.15 to 150 mc shall apply for the radiated radio noise test.

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4.5.2.2.10 Indicator magnetic effect. The indicator, not operating, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0, 45, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in the normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 gauss. The deflection of the compass at any of the specified positions shall not exceed 3 degrees. The test shall be repeated with the indicator operated at rated voltage.

4.5.2.2.11 Indicator thermal shock. The indicator shall be subjected to 8 cycles of exposure to water baths at +85° ± 2°C and +5° ± 2°C without evidence of moisture penetration or damage to the coating or enclosure. Each cycle of the test shall consist of immersing the indicator in the water bath at +85°C for a period of 30 minutes and then within 5 seconds of removal from the bath, the indicator shall be immersed for a period of 30 minutes in the other bath maintained at +5°C. This cycle shall be repeated continuously, 1 cycle following the other, until 8 cycles have been completed. Following this test, the indicator shall be subjected to the leak test, (4.5.1.6). No indicator leakage shall occur as a result of this test.

4.5.2.2.12 Transmitter substitute fuel. The transmitter shall be subjected to the friction and scale error test with fuel conforming to MIL-F-5572, and the specific gravity shall be 0.700 ± 0.005 at room temperature. The scale errors shall not exceed the tolerances specified in the detail specification.

4.5.2.2.13 Transmitter pressure drop. The pressure drop, with the transmitter movable element blocked in the zero flow position, shall be measured by any suitable method and shall not exceed the value specified in the detail specification.

4.5.2.2.14 Endurance. The indicator system shall be subject to 100,000 applications of flow to produce an indication from 0 to 80 ± 5 percent of the range of the system and back to zero at a rate of 15 ± 10 cpm. If the elapsed time of the endurance test is less than 1,200 hours when the specified number of cycles has been completed, the operation shall be continued until a total of 1,200 hours of operation has been completed. The friction and scale error test shall be conducted not less than 1 hour after the completion of the endurance cycling, and the errors shall not exceed the tolerances specified in the detail specification.

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4.5.2.2.15 Vacuum tube replacement. The indicator and intermediate device shall be electrically connected to each other and to a suitable master transmitter. The master transmitter shall be adjusted to a value such that the indicator reads approximately 2,000 pounds per hour (pph), or to any other points as desired by the activity responsible for qualification. Each vacuum tube or sealed unit containing vacuum tubes employed in the system, shall be removed individually and replaced consecutively with 6 tubes or units of the same type designation taken at random from the stock. The difference in reading obtained with the original tube and any of the stock replacement tubes or units shall not exceed 0.3 percent of full scale travel (320 degrees).

4.5.2.2.16 Indicator response time. The time required for the indicator to change from zero to full scale indication shall not exceed the maximum time specified in the detail specification.

4.5.3 Rejection. When sampling tests are specified on a number of items that are selected as representative of a certain lot, and one or more of this number fails to meet the specified test(s), acceptance of all items shall be withheld until the extent and cause of failure is determined. For operational reasons, individual tests may be continued pending investigation of a sampling test failure, but the final acceptance of the product is contingent upon the overall conformance of the product to specification requirements. If the investigation indicates that the defects may exist on items previously accepted, full particulars concerning the defect(s) found, including recommendations for correction, will be furnished to the procuring activity.

*4.6 Inspection and test. Test methods of preservation and packaging shall be accomplished in accordance with Section 4 of Specification MIL-P-116 to insure compliance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

*5.1.1 Level A. (Reference MIL-STD-794 for definition of levels of preservation and packaging.) Indicator Fuel Flowmeter shall be preserved and packaged in quantity unit packs of one each in accordance with Method III of Specification MIL-P-116, provided water-vaporproof caps are utilized on electrical connectors. Otherwise, Method ICi of MIL-P-116 shall apply. Sufficient cushioning of type and density shall be used to insure shock transmission does not exceed peak values in Gs established for items when completed packs are subjected to the rough handling drop test of MIL-P-116.

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*5.1.2 Level C. Same as Level A above.

*5.2 Packing.

*5.2.1 Level A. (Reference MIL-STD-794 for definitions of levels of packing.) Indicator Fuel Flowmeter shall be packed in shipping containers conforming to PPP-B-636, Weather-Resistant Class, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping containers shall be of uniformed shape, size, of minimum tare and cube, consistant with the protection required.

*5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in domestic class shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above shall apply.

*5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to the applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this specification.

5.3 Marking. Interior and exterior containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Instruments covered by this specification are intended for use in providing remote indication of fuel flow for aircraft engines.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City ALC/MMEO Tinker Air Force Base, Oklahoma 73145 and information pertaining to qualification of products may be obtained from that activity.

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6.4 Definitions.

6.4.1 Double amplitude. Whenever the term "double amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.4.2 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.5 The preferred list of electron tubes consists of those listed in Standard MIL-STD-200.

6.6 Changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 71

Project Number:
6620-F090

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